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CMPUT 365: Introduction to Reinforcement Learning

**Worksheet: Monte Carlo Methods**

**Question 1.** In Policy Iteration, we used dynamic programming for the policy evaluation step, to compute $v_\pi$. Monte Carlo ES is a Generalized Policy Iteration (GPI) algorithm, that does not do a full policy evaluation step, before greedifying. How might you modify Monte Carlo ES, to do (more) complete policy evaluation steps before greedifying?

**Question 2.** Off-policy Monte Carlo prediction allows us to use sample trajectories to estimate the value function for a policy that may be different than the one used to generate the data. Consider the following MDP, with two states $B$ and $C$, with 1 action in state $B$ and two actions in state $C$, with $\gamma = 1.0$. In state $C$ both actions transition to the terminating state with $A = 1$ following the blue path to receive a reward $R = 1$ and $A = 2$ following the green path to receive a reward $R = 10$. Assume the target policy $\pi$ has $\pi(A = 1|C) = 0.9$ and $\pi(A = 2|C) = 0.1$, and that the behaviour policy $b$ has $b(A = 1|C) = 0.25$ and $b(A = 2|C) = 0.75$.

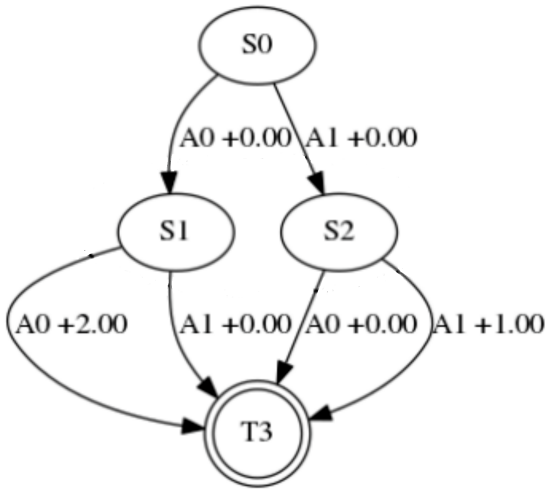
(a) What are the true values $v_\pi$?

(b) Imagine you got to execute $\pi$ in the environment for one episode, and observed the episode trajectory $S_0 = B, A_0 = 1, R_1 = 1, S_1 = C, A_1 = 1, R_2 = 1$. What is the return for $B$ for this episode? Additionally, what are the value estimates $V_\pi$, using this one episode with Monte Carlo updates?

(c) But, you do not actually get to execute $\pi$; the agent follows the behaviour policy $b$. Instead, you get one episode when following b, and observed the episode trajectory $S_0 = B, A_0 = 1, R_1 = 1, S_1 = C, A_1 = 2, R_2 = 10$. What is the return for $B$ for this episode? Notice that this is a return for the behaviour policy, and using it with Monte Carlo updates (without importance sampling ratios) would give you value estimates for $b$.

(d) But, we do not actually want to estimate the values for behaviour $b$, we want to estimates the values for $\pi$. So, we need to use importance sampling ratios for this return. What is the return for B using this episode, but now with importance sampling ratios? Additionally, what is the resulting value estimate for $V_\pi$ using this return?

**Question 3.** Consider the three-state MDP below with terminal state $T_3$ and $\gamma = 1$. Suppose you observe three episodes: $\{S_0, S_1, T_3\}$ with a return of 2, $S_0, S_1, T_3$ with a return of 2, $S_0, S_2, T_3$ with a return of 1. What is the (every-visit) Monte-Carlo estimator of the value for each of the states $S_0, S_1, S_2$? How would the Monte-Carlo estimates change if $r(S_0, A_1, S_1) = +1.00$?




**Question 4.** Consider the three-state MDP from Question 3, but let $r(S_1, A_1, T_3) = +1.00$ and $r(S_2, A_0, T_3) = +2.00$. You observe the following episodes: $\{S_0, S_1, T_3\}$ with a return of 2, $\{S_0, S_1, T_3\}$ with a return of 2, $\{S_0, S_1, T_3\}$ with a return of 1, $\{S_0, S_2, T_3\}$ with a return of 1. What is the (every-visit) Monte-Carlo estimator of the value for each of the states $S_0, S_1, S_2$?

**Question 5.** Suppose you would like to estimate the expected outcome for $3X^2$ where $X$ is the outcome of a fair die. You run an experiment and observe the following sequence of rolls

$$1, 1, 6, 3, 4.$$

How would you use the above data to estimate the expected value $\mathbb{E}[3X^2]$ for a fair die if the above rolls where observed from a loaded die with $P(X = 1) = 1/2$ and $P(X = i) = 1/10$ for $i = 2, 3, 4, 5, 6$?

**Question 6.** Off-policy Monte Carlo prediction allows us to use sample trajectories to estimate the value function for a policy that may be different than the one used to generate the data. Assume the behaviour policy is uniform random: it takes each action with equal probability in each state. The environment has two possible states $\{1, 2\}$ and two possible actions in each state $\{1, 2\}$, with $\gamma = 1.0$. Suppose we collect the following episode, by following the behaviour policy

$$S_0 = 1, A_0 = 2, R_1 = 3, S_1 = 2, A_1 = 1, R_2 = -4,$$

But, we want to evaluate target policy $\pi$, which selects action 1 with 0.3 probability in each state. Give an estimate of $V_\pi(S = 1)$ and $V_\pi(S = 2)$, using the episode generated under $b$.

**Question 7.** Exercise 5.2 from Sutton & Barto (2018).

**Question 8.** Exercise 5.3 from Sutton & Barto (2018).

**Question 9.** Exercise 5.4 from Sutton & Barto (2018).

**Question 10.** Exercise 5.5 from Sutton & Barto (2018).

**Question 11.** Exercise 5.11 from Sutton & Barto (2018).